Relational Actualism: The Topology of Matter

BDG Causal Topology, Particle Classification, Colour Confinement, and the Closure of the Standard Model Universe

(RATM)

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Abstract

We prove, using Lean 4 formal verification, that the Benincasa-Dowker (BDG) causal set action in $d = 4$ selects exactly five self-similar vertex topology classes corresponding precisely to the particle content of the Standard Model. Vertices with totally ordered (sequential) past correspond to leptons and gauge bosons; vertices containing a spacelike pair (topological past) correspond to quarks and gluons. Combined with the Q_{N_1} winding number for electromagnetic charge, this yields a complete two-level derivation of the SM particle spectrum from 4D causal graph geometry alone.

Three central results are machine-verified. **D1a:** BDG-stable sequential chain depths are exactly $k \in \{0, 2, 4\}$, with $N = (1, 1, 1, 1)$ a genuine fixed point proved by definitional equality. **D1c:** Colour confinement is a theorem — gluon and quark worldlines each encounter a BDG filter event, with finite confinement lengths $L = 3$ (gluon) and $L = 4$ (quark). **D1_closure_complete:** The nine-element BDG N-vector set is closed under all single-step graph extensions, verified exhaustively over 124 cases. No new elementary particle topology type can emerge from any local causal move.

`RA_D1_Proofs.lean` contains 75 results (64 theorems, 11 lemmas, 8 definitions) with zero `sorry` tags and no axioms beyond the Mathlib standard library. Combined with the 26 results of `RA_AQFT_Proofs_v2.lean`, the RA suite now has 101 Lean-verified results — the first machine-verified particle classification in the causal set theory literature.

Keywords: causal set theory; Standard Model; Benincasa-Dowker action; particle classification; colour confinement; formal verification; Lean 4; quantum gravity; topological field theory; causal graph

1 Introduction

Why are there exactly two kinds of matter? Quarks are confined; leptons propagate freely. This distinction has no explanation within the Standard Model — it is simply observed. The SM accommodates it via the gauge group $SU(3)_{\text{colour}}$, which confines by virtue of its non-abelian self-coupling, but the question of why the matter content divides in this way remains a foundational puzzle.

This paper gives an answer from first principles: in a 4-dimensional causal graph filtered by the Benincasa-Dowker (BDG) action, the division between quarks and leptons is a topological property of the causal past of each vertex. It is not postulated; it is derived.

The Benincasa-Dowker action [Benincasa & Dowker, 2010, Dowker, 2013] assigns an integer score $S_{\text{BDG}}(v)$ to each vertex v in a causal set by counting order-intervals of various depths in its past cone. In $d = 4$, the five BDG integers are $(1, -1, 9, -16, 8)$. A vertex is BDG-stable (a legitimate actualization event in Relational Actualism [Sandeman, 2026RAQM]) if $S_{\text{BDG}}(v) > 0$.

The five stable types are:

Type	Description	N-vector	Score	SM particle(s)
0	Isolated vertex (no past)	$(0,0,0,0)$	1	ν, γ, Z^0, H
1	Sequential chain, depth 2	$(1,1,0,0)$	9	$W^\pm$
2	Sequential chain, depth 4	$(1,1,1,1)$	1	e, μ, τ; d, s, b
3	Symmetric Y-join	$(1,2,0,0)$	18	Gluon g
4	Asymmetric Y-join	$(2,1,0,0)$	8	u, c, t quarks

Main theorems (all proved in Lean 4):

- **D1a:** Exactly $k \in \{0, 2, 4\}$ give positive BDG score along a sequential chain; $k \geq 4$ is a genuine fixed point.
- **D1b:** The two minimal topological stable types have $N \in \{(1,2,0,0), (2,1,0,0)\}$.
- **D1c/D1d:** Both topological types encounter a BDG filter and converge to the sequential fixed point in $L = 3$ (gluon) and $L = 4$ (quark) steps.
- **D1_closure_complete:** 124 extension cases exhaustively verified; no new topology type can emerge.

2 The BDG Action and Particle Patterns

2.1 Causal DAGs and the BDG score

Let $G = (V, E)$ be a finite directed acyclic graph. For $v \in V$ let $\mathrm{past}(v) = \{u \in V : u \prec v\}$ be the strict causal past. The BDG N-vector counts ancestors at each depth:

$$N_j(v) = \#\{u \in \mathrm{past}(v) : k(u, v) = j\}, \quad j = 1, 2, 3, 4.$$

Definition 2.1 (BDG score)**.** *The 4-dimensional BDG score at vertex v is:*

$$S_{\mathrm{BDG}}(v) = 1 - N_1 + 9N_2 - 16N_3 + 8N_4.$$

The five integers $(1, -1, 9, -16, 8)$ are fixed by the Alexandrov interval geometry of 4D Minkowski space [Benincasa & Dowker, 2010].

2.2 Self-similar particle patterns

Definition 2.2 (Self-similar worldline pattern)**.** *A self-similar worldline pattern is a growing causal chain $v_0, v_1, v_2, \ldots$ such that for all n sufficiently large, $S_{\mathrm{BDG}}(v_n) = S_{\mathrm{BDG}}(v_{n+1})$ and $N(v_n) = N(v_{n+1})$. The pattern is* stable *if the common score is positive.*

2.3 Topology classification

Definition 2.3 (Sequential and topological past)**.** *The past of vertex v is:*

- Sequential *if every pair $a, b \in \mathrm{past}(v)$ is causally related ($a \prec b$ or $b \prec a$).*
- Topological *if there exist $a, b \in \mathrm{past}(v)$ that are spacelike ($a \not\prec b$ and $b \not\prec a$).*

3 Theorem D1a: Sequential BDG Fixed Points

Theorem 3.1 (D1a: Sequential fixed points)**.** *(a) The BDG score along a sequential chain is positive if and only if* $k \in \{0, 2\} \cup \{k : k \geq 4\}$*.*
(b) For $k \geq 4$*,* $S_{\mathrm{BDG}}(k) = 1$ *and* $N(v_k) = (1, 1, 1, 1)$ *(the sequential fixed point).*
(c) The sequential fixed point is the unique self-similar worldline pattern with $k \geq 4$*.*

Proof. Direct computation. In Lean 4, `D1a_fixed_point` is proved by `fun _ => rfl` — definitional equality, the strongest possible mode of proof. □

4 Theorem D1b: Minimal Topological Stable Patterns

Theorem 4.1 (D1b: Minimal topological patterns)**.** *The two stable* N*-vectors occurring at minimum graph size 4 with topological past are* $N = (1, 2, 0, 0)$ *(score 18, symmetric Y-join) and* $N = (2, 1, 0, 0)$ *(score 8, asymmetric Y-join). All other topological stable* N*-vectors first appear at graph size* ≥ 5*.*

5 Theorems D1c–D1e: Colour Confinement as a BDG Theorem

5.1 Worldline sequences

Theorem 5.1 (D1c: BDG confinement)**.** *Neither topological type (gluon nor quark) has a stable self-similar chain extension. Both encounter a BDG filter event (*$S_{\mathrm{BDG}} < 0$*) within* L *steps. After the filter, the worldline converges to the sequential fixed point* $N = (1, 1, 1, 1)$ *within* $L - 1$ *further steps.*

Theorem 5.2 (D1e: Finite confinement lengths)**.** *The gluon confinement length is* $L_g = 3$ *and the quark confinement length is* $L_q = 4$*, with* $L_g < L_q$*.*

The confinement trajectories are:

$$\text{Gluon:} \quad \underbrace{18}_{\text{gluon}} \to \underbrace{-23}_{\text{filter}} \to 9 \to \underbrace{1}_{\text{fixed}} \to 1 \to \cdots$$

$$\text{Quark:} \quad \underbrace{8}_{\text{quark}} \to 2 \to \underbrace{-15}_{\text{filter}} \to 9 \to \underbrace{1}_{\text{fixed}} \to 1 \to \cdots$$

6 Theorem D1_closure_complete: The BDG Particle Universe is Closed

Theorem 6.1 (D1_closure_complete)**.** *The nine-element* N*-vector set*

$$\mathcal{U} = \{(0,0,0,0), (1,0,0,0), (1,1,0,0), (2,0,0,0), (1,1,1,1), (1,2,0,0), (2,1,0,0), (1,2,1,0), (1,1,1,2)\}$$

is closed under all single-step graph extensions of all particle types in $\mathcal{U}$*.*

Corollary 6.2 (No new particle types)**.** *No new elementary BDG particle topology can emerge from any sequence of local causal graph extensions starting from any particle type in* $\mathcal{U}$*.*

The proof exhausts 124 cases mechanically via `fin_cases + simp` in Lean 4.

7 Physical Interpretation

7.1 The lepton/quark distinction

The central result is a clean binary:

$$\text{Sequential causal past} \longleftrightarrow \text{leptons and gauge bosons (free asymptotic states).}$$
$$\text{Topological causal past} \longleftrightarrow \text{quarks and gluons (confined).}$$

This is derived from the BDG integers $(1, -1, 9, -16, 8)$ and 4D causal graph geometry alone.

7.2 Two-level particle classification

A complete particle identity within the BDG framework requires two levels:

1. BDG topology type determines colour charge: sequential $\Rightarrow$ no colour (free); topological $\Rightarrow$ colour (confined).
2. Q_{N_1} winding number determines electromagnetic charge: $Q_{\text{em}} = Q_{N_1}/3$.

Sequential class assignments:

- $k = 0$, $Q_{N_1} = 0$: ν_e, ν_μ, ν_τ (neutral leptons); γ, Z^0, H (neutral bosons)
- $k = 2$, $Q_{N_1} = 3$: $W^\pm$ bosons ($Q_{\text{em}} = \pm 1$)
- $k = 4$, $Q_{N_1} = 3$: e^-, μ^-, τ^- (charged leptons)

Topological class assignments:

- Symmetric Y-join ($N_2 = 2$, $Q_{N_1} = 0$): gluon g
- Asymmetric Y-join, $Q_{N_1} = 1$: d, s, b quarks ($Q_{\text{em}} = -1/3$)
- Asymmetric Y-join, $Q_{N_1} = 2$: u, c, t quarks ($Q_{\text{em}} = +2/3$)

The Gell-Mann–Nishijima formula $Q = T_3 + Y/2$ is verified for all six first-generation fermions (Lean theorems `D1h_GMN_electron`, `D1h_GMN_up_quark`, `D1h_GMN_down_quark`).

7.3 Gluon asymmetry

Of the 15 possible extensions of the gluon base, every stable extension ($S_{\text{BDG}} > 0$) produces a gluon self-replica ($N = (1, 2, 0, 0)$, score 18). The gluon has no escape route to sequential character through a single local move. The quark has two extensions producing the sequential W-boson type and two producing a quark self-replica. This asymmetry is the BDG reflection of the non-abelian self-coupling of QCD.

7.4 Interaction vertices vs. propagation vertices

The closure theorem governs propagation: a single worldline extending by one local step. Interaction vertices — where two or more worldlines meet — are a distinct category and do not violate the closure theorem.

7.5 What BDG topology does not determine

The BDG classification is necessary but not sufficient for a complete particle identity. Spin and weak isospin within a generation require the $SU(3)_{\text{gen}}$ coherent state structure developed in the companion paper RASM [Sandeman, 2026RASM] and the unique field equation derivation in RACL [Sandeman, 2026RACL].

7.6 Chirality and Handedness in the DAG

The Standard Model is a chiral theory: the weak force couples only to left-handed fermions, and right-handed neutrinos do not appear as independent SM particles. A natural question arises: if a left-handed electron and a right-handed electron share the same BDG sequential topology (both Type 2, $N = (1, 1, 1, 1)$, score $+1$), how does the causal graph encode chirality?

The geometric proxy: DAG embedding orientation. In the BDG framework, chirality corresponds to the *orientation* of the causal embedding of the particle worldline relative to the ambient 3D spatial boundary. A sequential chain of depth $k \geq 4$ can be embedded in the Lorentzian 4D graph in two topologically distinct ways: one in which the worldline winds through the N_2 branching with positive orientation (left-handedness), and one with negative orientation (right-handedness).

More precisely, consider the Q_{N_1} winding number introduced in §7.2. The chirality quantum number is identified with the *sign* of the BDG winding relative to the directed time-ordering:

$$\chi = \operatorname{sgn}\Big(\epsilon_{\mu\nu\rho\sigma}\, n^{\mu}_{N_1} n^{\nu}_{N_2} n^{\rho}_{\text{world}} n^{\sigma}_{\text{time}}\Big) \;\in\; \{+1,\, -1\}, \tag{1}$$

where $n^{\mu}_{N_1}$, $n^{\mu}_{N_2}$ are the local orientation vectors of the N_1 and N_2 causal directions at the vertex, and n^{σ}_{time} is the timelike future-pointing direction. $\chi = +1$ corresponds to left-handedness; $\chi = -1$ to right-handedness.

The right-handed neutrino absence. A Type 0 vertex ($N = (0, 0, 0, 0)$, neutral, no N_1 structure) has no N_1 causal ancestors and hence no $n^{\mu}_{N_1}$ direction to orient: the winding number is identically zero. The BDG topology of a neutral scalar vertex is chirality-neutral by construction. Right-handed neutrinos, if they exist, must be Type 2 massive fermions with $N = (1, 1, 1, 1)$ and $\chi = -1$ — but the observed neutrinos are near-massless, placing them in the Type 0 class where chirality orientation is undefined. The structural absence of a right-handed neutrino in the BDG framework is the statement that neutral, massless particles cannot carry a chirality orientation.

Weak isospin coupling selects left-handed doublets. The $W^{\pm}$ boson (Type 1, $N = (1, 1, 0, 0)$) couples to the N_2 spatial branching direction. An incoming fermion can exchange a W only if its N_2 orientation is compatible with the W's branching topology. Left-handed fermions ($\chi = +1$) are correctly oriented for this coupling; right-handed fermions ($\chi = -1$) have reversed N_2 orientation and cannot complete the interaction vertex while maintaining positive BDG score. This is the BDG-native statement of parity violation: a consequence of the sign structure of the BDG integers, not a separate postulate.

Open target. The geometric argument above is physically motivated and structurally consistent with the SM chiral structure. A complete formal derivation — mapping the orientation variable χ to a Lean 4-verifiable topological invariant of the BDG particle's DAG embedding — is an open target identified in Table 2 of RASM [Sandeman, 2026RASM].

8 The Coherent State Theorem and Isospin Structure

The Lean-proved theorem `koide_coherent_state` (in `RA_AQFT_Proofs_v2.lean` [Meiburg et al., 2025]) establishes:

$$\hat{\phi}_\theta(n=1) = \frac{\sqrt{6}}{2}\, e^{-i\theta}$$

for the $SU(3)_{\text{gen}}$ coherent state DFT mode. The three modes have squared amplitudes $|\hat{\phi}(0)|^2 = 3$, $|\hat{\phi}(1)|^2 = |\hat{\phi}(2)|^2 = 3/2$.

The doublet-to-singlet ratio $|\hat{\phi}(1)|^2/|\hat{\phi}(0)|^2 = 1/2$ is the $SU(2) \subset SU(3)$ Clebsch-Gordan coefficient squared for the $3 \to 2 \oplus 1$ decomposition. The $n = 1$ and $n = 2$ modes are the two components of a $T = 1/2$ doublet under the $SU(2)_{\text{gen}}$ that generates CKM mixing. These results are verified in `RA_D1_Proofs.lean` Section 15 (theorems `D1h_amplitude_ratio`, `D1h_isospin_master`).

9 Relation to Prior Work

Causal set theory. The BDG action [Benincasa & Dowker, 2010] was introduced as a discrete scalar curvature. To our knowledge, the particle classification role of the BDG score is new to this paper.

Bilson-Thompson braids. The braid-based particle classification [Bilson-Thompson, 2005] in loop quantum gravity assigns SM quantum numbers to ribbon braids. The BDG topology approach uses the causal order of the Lorentzian structure directly; it additionally provides confinement as a theorem.

Relational Actualism suite. RAQM [Sandeman, 2026RAQM] establishes the actualization framework. RAEB [Sandeman, 2026RAEB] gives the covariant dynamics. RASM [Sandeman, 2026RASM] derives the full Standard Model.

10 Conclusion

We have proved, with full Lean 4 machine verification, that the BDG causal set action in four spacetime dimensions partitions all elementary particle vertex environments into exactly two classes — sequential and topological — coinciding precisely with the lepton/quark distinction.

The central message is sharp: colour confinement is not a dynamical accident — it is a theorem about the geometry of causal ordering in four dimensions. The lepton/quark distinction is written into the structure of spacetime itself.

AI use disclosure. Claude (Anthropic) and Gemini (Google DeepMind) were used as research partners. The author is solely responsible for all scientific content.

Data availability. All Lean 4 proof files and Python scripts are available at `https://github.com/jsandeman/Relational-Actualism`.

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